1. **Develop a program to blink 5 LEDs back and forth.**

```
int ledPins[] = {2, 3, 4, 5, 6};
int numLeds = 5;

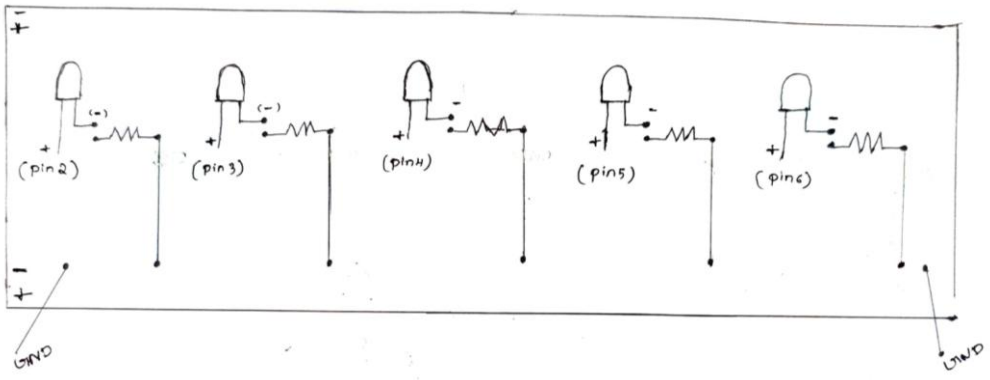
void setup() {
  // Set all LED pins as output
  for (int i = 0; i < numLeds; i++) {
    pinMode(ledPins[i], OUTPUT);
  }
}

void loop() {
  // Forward direction (left to right)
  for (int i = 0; i < numLeds; i++) {
    turnOnOnly(i);
    delay(150);
  }

  // Backward direction (right to left)
  for (int i = numLeds - 2; i > 0; i--) {
    turnOnOnly(i);
    delay(150);
  }
}

// Function to turn on one LED and turn off the rest
void turnOnOnly(int ledIndex) {
  for (int i = 0; i < numLeds; i++) {
    digitalWrite(ledPins[i], i == ledIndex ? HIGH : LOW);
  }
}
```

OUTPUT



2. Develop a program to interface a relay with Arduino board.

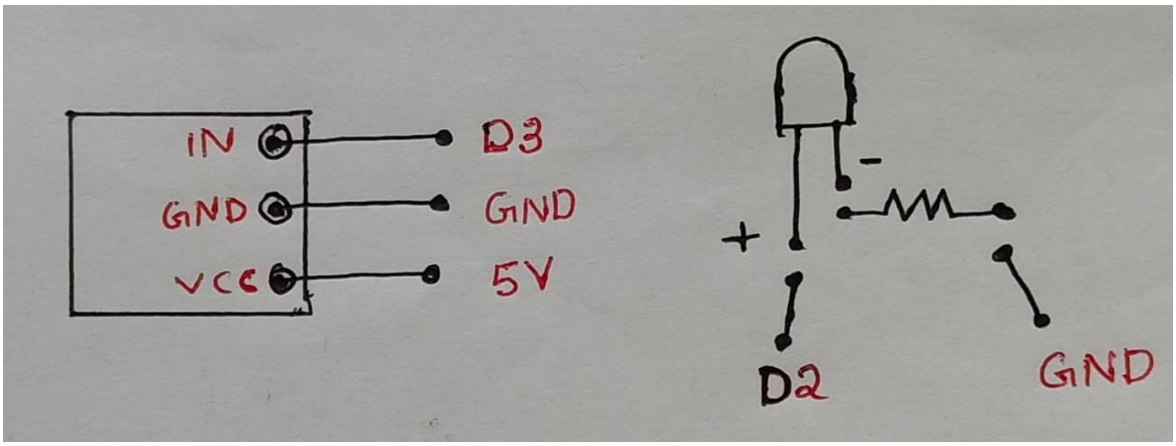
```
const int ledPin = 2;
const int relayPin = 3;

void setup() {
  pinMode(ledPin, OUTPUT);
  pinMode(relayPin, OUTPUT);
}

void loop() {
  // Turn ON LED and Relay
  digitalWrite(ledPin, HIGH);
  digitalWrite(relayPin, LOW);
  delay(1000);

  // Turn OFF LED and Relay
  digitalWrite(ledPin, LOW);
  digitalWrite(relayPin, HIGH);
  delay(1000); // OFF for 1 second
}
```

OUTPUT



3. Develop a program to deploy an intrusion detection system using Ultrasonic and sound sensors.

```
const int trigPin = 9;
const int echoPin = 10;
const int buzzer = 7;

long duration;
int distance;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(buzzer, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  // Send ultrasonic pulse
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Read echo response
  duration = pulseIn(echoPin, HIGH);
```

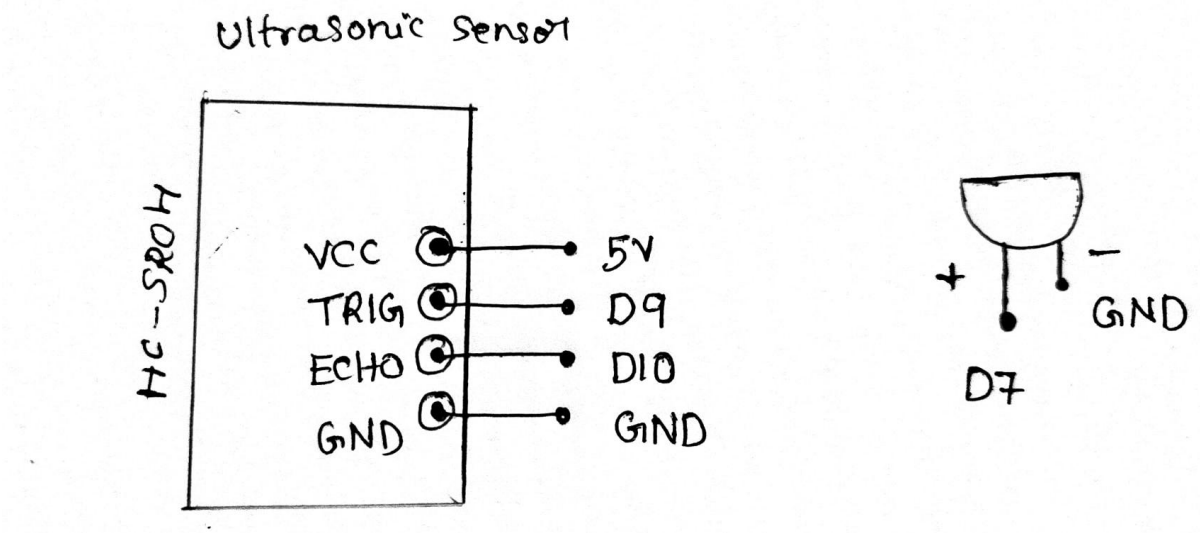
```
  // Calculate distance in cm
  distance = duration * 0.034 / 2;

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  // Intrusion detection logic
  if (distance < 20) { // If object is closer than 20 cm
    digitalWrite(buzzer, HIGH); // Trigger alarm
  } else {
    digitalWrite(buzzer, LOW);  // No intrusion
  }

  delay(500);
}
```

OUTPUT



3. Develop a program to deploy an intrusion detection system using Ultrasonic and sound sensors.

```
const int trigPin = 9;
const int echoPin = 10;
const int soundPin = 8;
const int buzzer = 7;

long duration;
int distance;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(soundPin, INPUT);
  pinMode(buzzer, OUTPUT);

  Serial.begin(9600);
}

void loop() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);

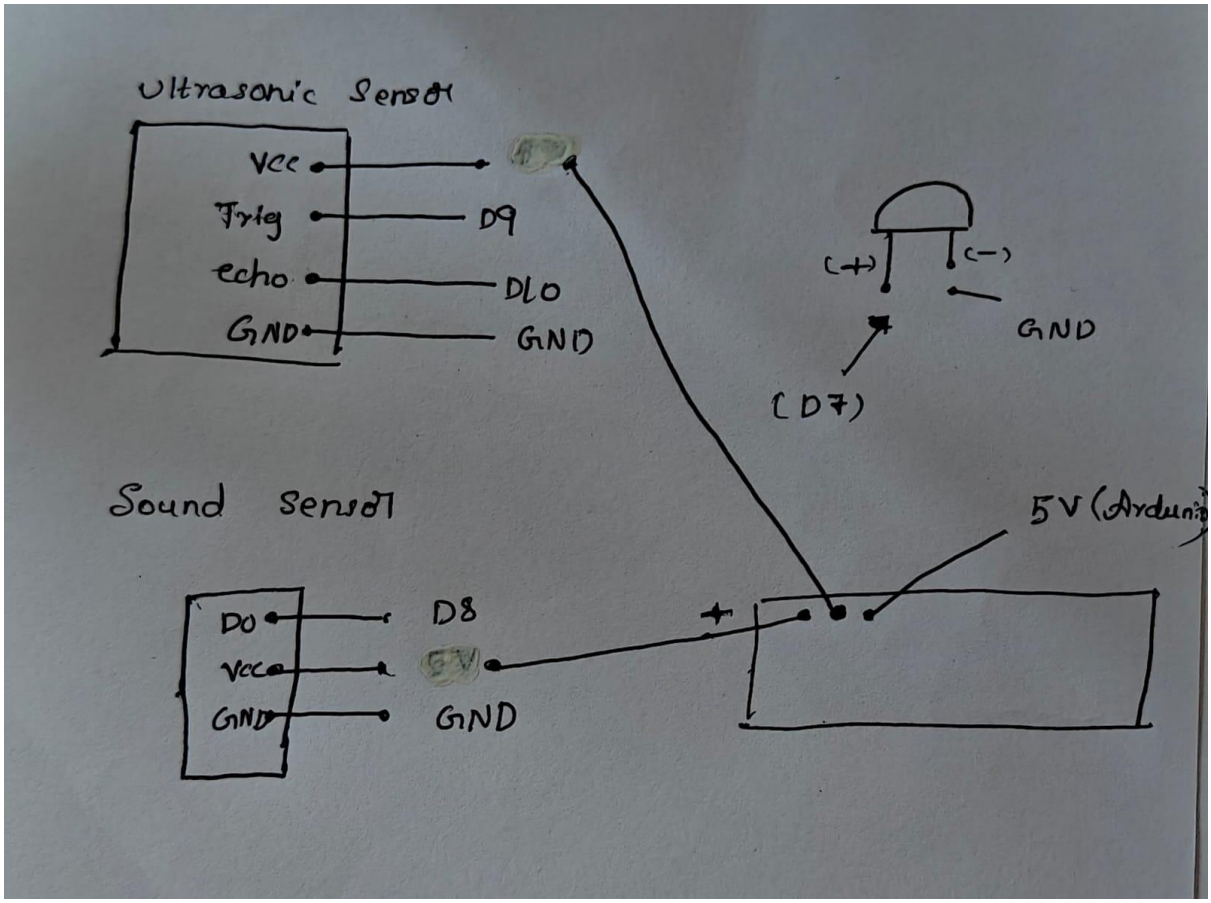
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH);
```

```
  // Calculate distance in cm
  distance = duration * 0.034 / 2;
  int soundState = digitalRead(soundPin);

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.print(" cm");

  if (soundState == LOW) {
    Serial.println(" | Loud sound detected");
  } else {
    Serial.println(" | No loud sound");
  }
  if (distance < 20 || soundState == LOW) {
    digitalWrite(buzzer, HIGH);
  }
  else {
    digitalWrite(buzzer, LOW);
  }
  delay(500);
}
```

OUTPUT


Ultrasonic Sensor
Vcc
Trig
echo
GND
D9
D10
GND
(+)
(-)
GND
(D7)
Sound Sensor
5V (Arduin
DO
Vcc
GND
D8
GND

4. Develop a program to control a DC motor with Arduino board.

```
// Motor control pins
const int IN1 = 8;
const int IN2 = 9;
const int ENA = 10;  // Must be PWM-capable

void setup() {
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(ENA, OUTPUT);
}

void loop() {
  // Spin motor forward
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);

  // Increase speed gradually
  for (int speed = 0; speed <= 255; speed += 5) {
    analogWrite(ENA, speed);
    delay(20);
  }

  delay(1000);  // Run at full speed for a while

  // Decrease speed
  for (int speed = 255; speed >= 0; speed -= 5) {
    analogWrite(ENA, speed);
    delay(20);
```

```
  }

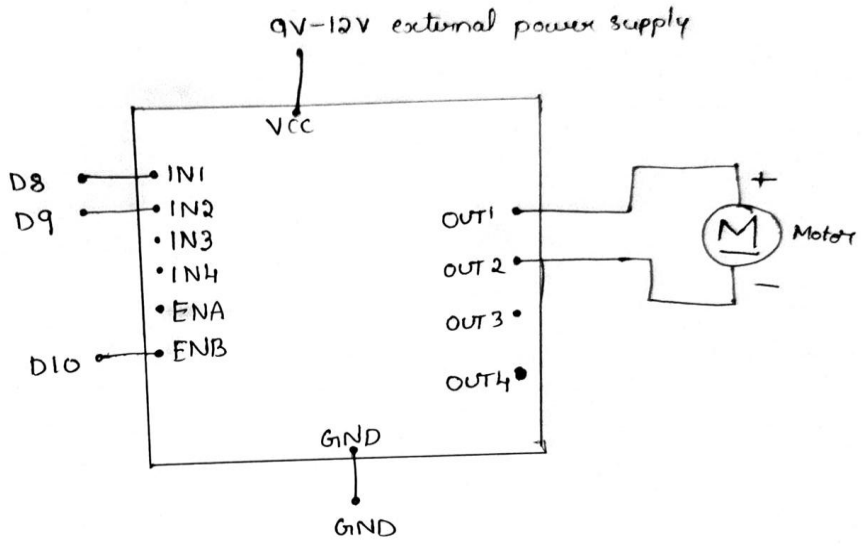
  delay(1000);  // Pause

  // Spin motor in reverse
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);

  // Full speed reverse
  analogWrite(ENA, 200);
  delay(2000);

  // Stop motor
  analogWrite(ENA, 0);
  delay(2000);
}
```

OUTPUT



5. Develop a program to deploy smart street light system using LDR sensor.

```
#define LDR_PIN 2      // Digital output from LDR module
#define LED_PIN 13     // LED as streetlight

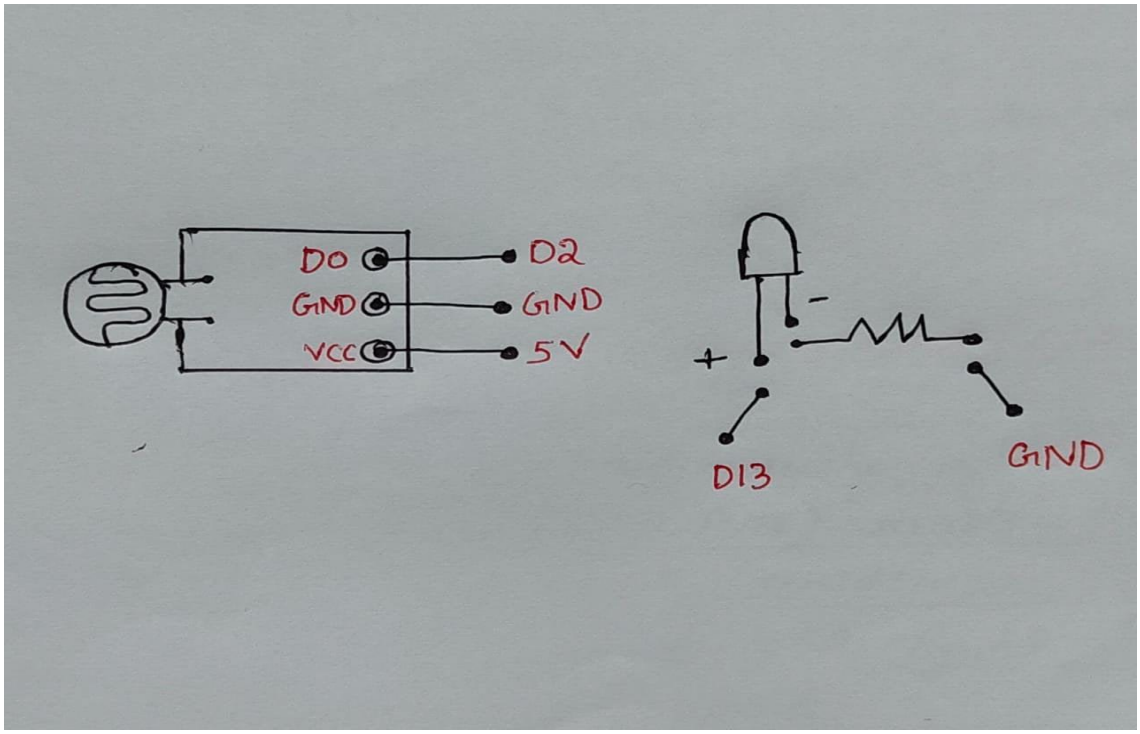
void setup() {
  pinMode(LDR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);
}

void loop() {
  int ldrState = digitalRead(LDR_PIN);

  if (ldrState == LOW) {
    // It's dark
    digitalWrite(LED_PIN, LOW);  // Turn ON street light
  } else {
    // It's bright
    digitalWrite(LED_PIN, HIGH);   // Turn OFF street light
  }

  delay(100);  // Small delay for stability
}
```

OUTPUT



6. Develop a program to classify dry and wet waste with the Moisture sensor (DHT22).

```
const int sensorPin = A0;    // Analog input pin for moisture sensor
const int ledPin = 8;        // LED to indicate wet waste
const int buzzerPin = 11;     // Buzzer for wet waste alert

int sensorValue = 0;
int threshold = 500;

void setup() {
  Serial.begin(9600);
  pinMode(ledPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);
}

void loop() {
  sensorValue = analogRead(sensorPin);
  Serial.print("Moisture Sensor Value: ");
  Serial.println(sensorValue);

  if (sensorValue < threshold) {
    Serial.println("Status: WET WASTE");
    digitalWrite(ledPin, HIGH);
    digitalWrite(buzzerPin, HIGH);
  } else {
    Serial.println("Status: DRY WASTE");
    digitalWrite(ledPin, LOW);
    digitalWrite(buzzerPin, LOW);
```

```
  }
  delay(1000);
}
```

OUTPUT

